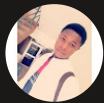




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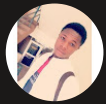
Connect a GitHub Repo with AWS



Ahmed Tetteh

```
Installed:
git-2.50.1-1.amzn2023.0.1.x86_64      git-core-2.50.1-1.amzn2023.0.1.x86_64  git-core-doc-2.50.1-1.amzn2023.0.1.noarch  perl-Error-1:0.17029-5.amzn2023.0.2.noarch
perl-File-Find-1.37-477.amzn2023.0.7.noarch  perl-Git-2.50.1-1.amzn2023.0.1.noarch  perl-TermReadKey-2.30-9.amzn2023.0.2.x86_64  perl-lib-0.65-477.amzn2023.0.7.x86_64

Complete!
[ec2-user@ip-172-31-47-94 ~]$ git --version
git version 2.50.1
[ec2-user@ip-172-31-47-94 ~]$
```



Introducing Today's Project!

In this project, I will demonstrate how to connect a Github Repo with AWS. I'm doing this project to learn how to simulate an actual working environment from code building to tracking changes in on Github.

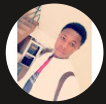
Key tools and concepts

Services I used were Amazon EC2, Git and Github actions, and remote SSH connection. Key concepts I learnt include how to initialize a local repo to track changes, link a local and remote repository, staging, pushing files to the remote repo and creating Personal Access Tokens (PATs) for authentication purposes.

Project reflection

This project took me approximately 1.5 hrs... The most challenging part was merging linking the local and remote repository without facing any merge issues. It was most rewarding to see my changes being tracked between my local and remote repository.

I did this project because its part of the CI/CD pipeline essential to writing and tracking multiple versions of the code.

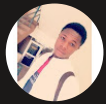


Ahmed Tetteh

NextWork Student

nextwork.org

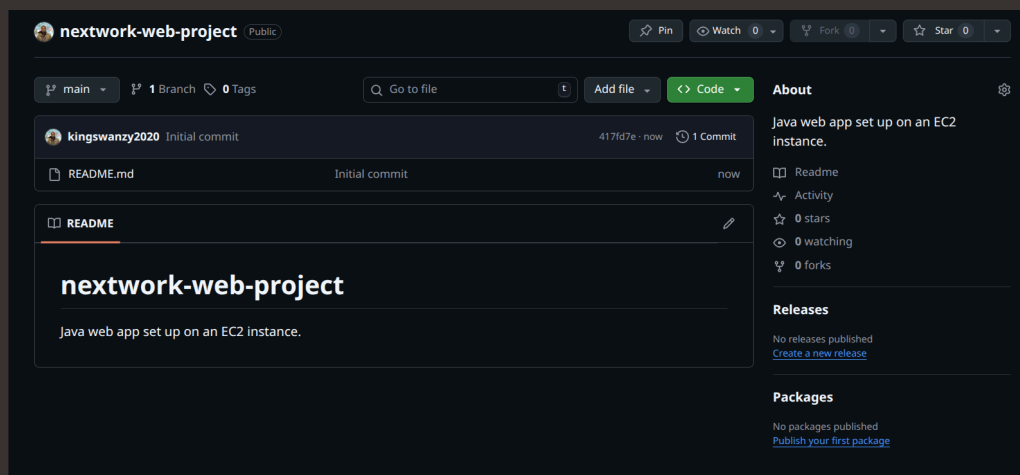
This project is part two of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project - AWS CodeArtifact on the 23rd Nov, 2025

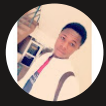


Git and GitHub

Git is a version control system used to track changes made to a file by taking snapshots of the file at a specific point in time. I installed Git using the commands "sudo dnf update -y" to update all existing software of the system, and "sudo dnf install git -y" to install git on the instance.

GitHub is the storage space on the web where all the code changes are tracked using Git. I'm using GitHub in this project to see upload and track all my code changes in a user-friendly interface.



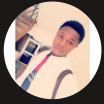


My local repository

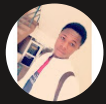
A Git repository is a storage space on the cloud that hosts all your projects tracked by Git, including the changes made and fosters collaboration between different engineers.

Git init is a command that initializes Git (version control system) for the current directory to track changes within it locally. I ran git init in the nextwork-web-project directory on my EC2 instance.

A branch in Git signifies different versions of the project, with the main or master branch being the head, and all other branches running parallel to it for different changes to be made and merged back to it at some point in the project. After running git init, the response from the terminal was "Initialized empty Git repository in /home/ec2-user/nextwork-web-project/.git/", meaning a directory has been initialized to track changes.



```
[ec2-user@ip-172-31-47-94 nextwork-web-project]$ pwd
/home/ec2-user/nextwork-web-project
[ec2-user@ip-172-31-47-94 nextwork-web-project]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ec2-user/nextwork-web-project/.git/
```



To push local changes to GitHub, I ran three commands

`git add`

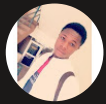
The first command I ran was "`git add .`"(the staging area). A staging area serves an intermediate space where you can store all your modified files and check them for any potential errors before they are committed.

`git commit`

The second command I ran was "`git commit -m`"(it saves or takes a snapshot of the staged files into the project history)... Using '`-m`' means add a message to the commit, describing what changed, and making it easier to follow for version review.

`git push`

The third command I ran was "`git push -u origin master`". Using '`-u`' means upstream. It tells git where to push the changes to, without defining the origin and branch in every push command.



Authentication

When I commit changes to GitHub, Git asks for my credentials because Git wants to authenticate who I am, and whether I have the right to push changes to my remote repository.

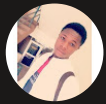
Local Git identity

Git needs my name and email to track the author information(username and email) of who made the commit.

Running git log showed me the commit history, including the author information, which in this case, was the system default username and Public IPV4 DNS.

```
[ec2-user@ip-172-31-47-94 nextwork-web-project]$ git log
commit 34952c8d78e1ea7b9c8caf39414653f5cf4c5530 (HEAD -> main, origin/main)
Author: EC2 Default User <ec2-user@ip-172-31-47-94.ap-northeast-1.compute.internal>
Date: Sat Nov 22 12:18:14 2025 +0000

    Updated index.jsp with new content
[ec2-user@ip-172-31-47-94 nextwork-web-project]$
```

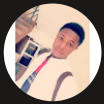


GitHub tokens

GitHub authentication failed when I entered my password because passwords through the terminal are insecure and can be intercepted over the internet via HTTPS. Hence, Personal Access Tokens(PATs) can be used instead to generate a string of random characters for the password.

A GitHub token is a unique string of randomly generated characters used as a password. Tokens can be used to control the permissions one has in the Github account when used. I'm using one in this project because passwords over internet are insecure, while Tokens provide a more fine grained control and security to the Github account.

I could set up a GitHub token by going to Settings > Developer settings > Personal access tokens > Tokens(classic) > Generate new token(classic).



GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

Generated for EC2 Instance Access. This is a part of NextWork's

What's this token for?

Expiration

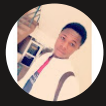
30 days (Dec 22, 2025)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows

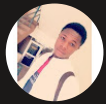


Making changes again

I wanted to see Git working in action, so I added a line of code in the body on my index.jsp file. I couldn't see the changes in my GitHub repo initially because the file hadn't been staged and committed to my local repo - then pushed to my Github repository.

I finally saw the changes in my GitHub repo after I pushed them from my local repo to my Github repository.

```
nextwork-web-project / src / main / webapp / index.jsp
kingswanzy2020 Add new line to index.jsp
Code Blame 14 lines (8 loc) · 295 Bytes
1 <html>
2
3 <body>
4
5 <h2>Hello King!</h2>
6
7 <p>This is my NextWork web application working!</p>
8 <p>I am writing this line using nano instead of an IDE.</p>
9 <p>If you see this line in Github, that means your latest changes are getting pushed to your cloud repo :o</p>
10
11
12 </body>
13
14 </html>
```

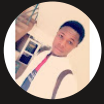


Setting up a README file

As a finishing touch to my GitHub repository, I added a README file, which is simply a project documentation file. I added a README file by creating one in my Github repository - then proceeded to document my project (what the project is about, the technologies I used and my setup).

My README is written in Markdown because it's a simple text language that can be easily read and converted to HTML to be displayed on webpages. Special characters can help you format text in Markdown, such as # for headers, ## subheaders,
 for line break in HTML style, and more

My README file has 6 sections that outline the overall project description, a brief intro about this part of the project, technologies used, setup, contact info and conclusion.



nextwork-web-project / README.md

kingswanzy2020 Update README with deployment pipeline details and Git tips

Preview Code Blame 66 Lines (47 loc) · 2.83 KB

Java Web App Deployment with AWS CI/CD

Welcome to this project combining Java web app development and AWS CI/CD tools!

Table of Contents

- [Introduction](#)
- [Technologies](#)
- [Setup](#)
- [Contact](#)
- [Conclusion](#)

Introduction

This project is used for an introduction to creating and deploying a Java-based web app using AWS, especially their CI/CD tools.

The deployment pipeline I'm building around the Java web app in this repository is invisible to the end-user, but makes a big impact by automating the software release processes. Am doing this project to learn how CI/CD pipelines can automate and speed up the software development life-cycle, and eventually, how AI can play a part in this process.



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